DATA VIZUALIZATION TOOLS

ASSIGNMENT 1 - DATA VIZUALIZATION LAB

Name - Manav Goel
PRN -24070126105
AIML B2

GITHUB PUBLISH LINK -

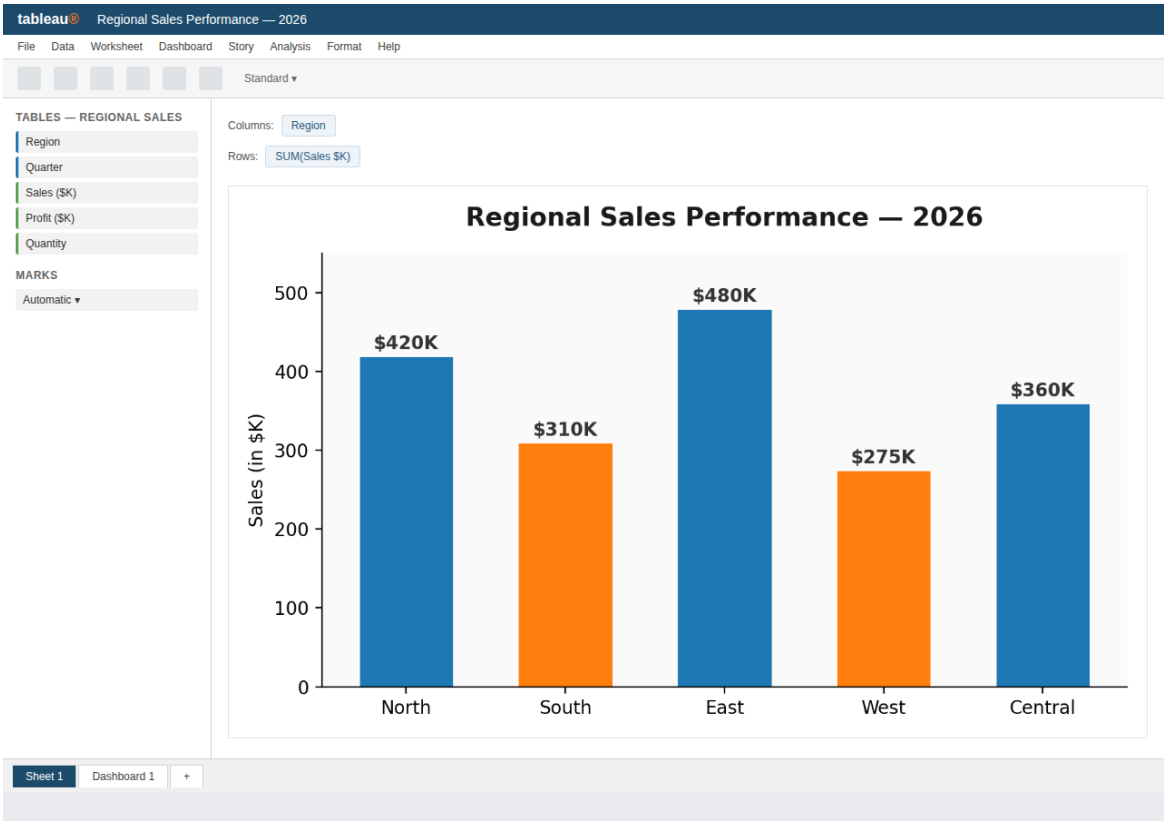
Dataset used - Regional Sales Performance, 2025

Data visualization is the bridge between raw numbers and real decisions. While Power BI often dominates the conversation, it's far from the only serious player in the field. In this post, we explore five other tools each with a distinct strength, and ideal use case with example on same data.

1. Tableau

Tableau is one of the most established names in business intelligence. Known for its powerful drag-and-drop interface, it lets analysts build rich, interactive dashboards without writing a single line of code — while still offering deep customization through calculated fields, parameters, and LOD (Level of Detail) expressions. It shines in enterprise settings where large, complex datasets need to be explored visually and shared across teams.

- Strengths: Enterprise-grade performance, huge visual gallery, strong community
- Cons : Licensing cost can be steep for small teams



Sample Tableau-style visualization — Regional Sales Performance (bar chart)

2. Looker Studio (formerly Google Data Studio)

Looker Studio is Google's free, cloud-native reporting tool. Its biggest advantage is how effortlessly it connects to the Google ecosystem Google Analytics, Google Ads, BigQuery, and Sheets making it a favorite among marketing and growth teams who want live, shareable reports without any infrastructure overhead.

- Strengths: Free, cloud-based, excellent for marketing dashboards
- Cons : Less powerful for very large or non-Google datasets



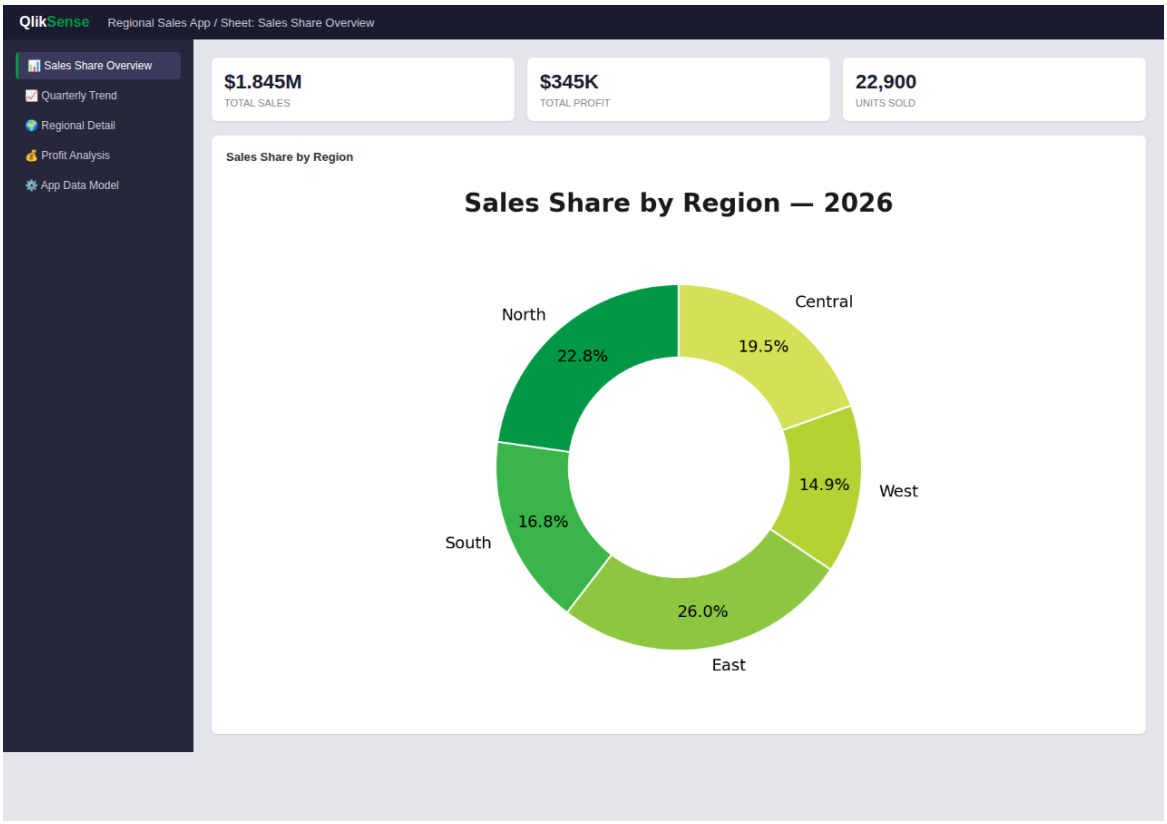
Sample Looker Studio-style visualization — Website Traffic Sources (line chart)

3. Qlik Sense

Qlik Sense takes a different approach with its associative data engine, which lets users explore relationships in data freely rather than following a fixed drill-down path. Selecting one data point instantly

highlights everything related to it across every chart on the page, which makes ad-hoc exploration feel remarkably intuitive.

- Strengths: Associative exploration, strong self-service analytics
- Cons : Smaller talent pool compared to Tableau users

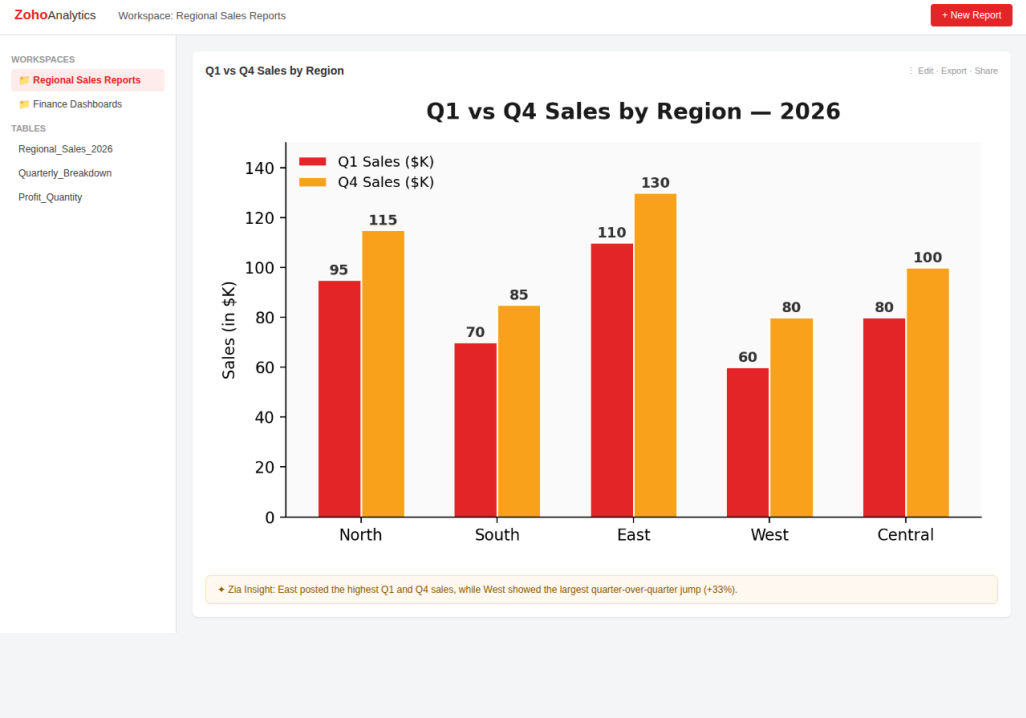


Revenue Share by Product Category (donut chart)

4. Zoho Analytics

Zoho Analytics is a self-service BI and analytics platform built to plug seamlessly into the wider Zoho ecosystem CRM, Books, Projects, and more while also connecting to external databases, cloud apps, and spreadsheets. It offers guided drag-and-drop report building, an AI-powered assistant (Zia) that answers questions in plain English, and affordable pricing that makes it especially popular with small and mid-sized businesses.

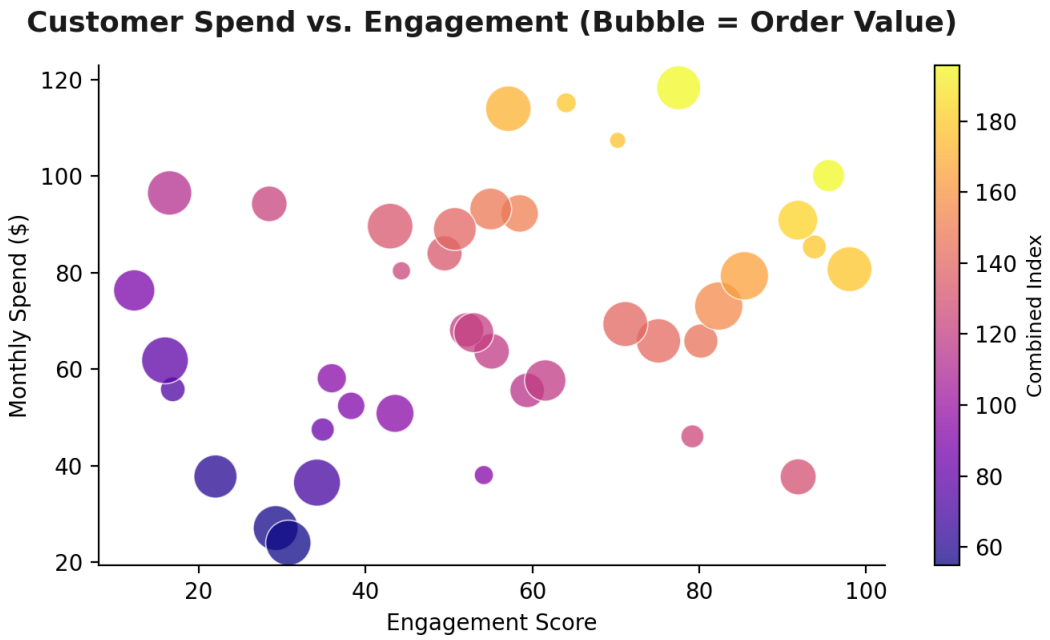
- Strengths: Fast setup, tight Zoho Suite integration, budget-friendly for SMBs
- Cons : Less depth for very large-scale enterprise data modeling



5. Plotly

Plotly bridges the gap between coding flexibility and ease of use. Available as a Python, R, or JavaScript , it produces publication-quality, fully interactive charts with hover tooltips, zooming, and panning built in by default a favorite among data scientists building analytical apps and notebooks.

- Strengths: Interactive by default, great for data science workflows
- Watch out for: Enterprise deployment (Dash Enterprise) comes at a cost



Sample Plotly-style visualization — Customer Spend vs. Engagement (bubble scatter chart)

Side-by-Side Comparison

Each tool optimizes for a different kind of user. The table below compares all five across the parameters that matter most when choosing a visualization tool.

Parameter	Tableau	Looker Studio	Qlik Sense	Zoho Analytics	Plotly
Ease of Use	Moderate — drag & drop, some learning curve	Very Easy — built for marketers	Moderate — associative UI takes practice	Easy — guided drag & drop, AI assistant (Zia)	Moderate — Python/R/JS knowledge helps
Cost	Paid (Free public version limited)	Free	Paid (Freemium tier available)	Paid (Free tier for small teams)	Free (open-source) + paid Enterprise/Cloud
Best For	Enterprise BI & executive dashboards	Marketing & Google-ecosystem reporting	Associative, self-service analytics	SMBs & Zoho-ecosystem reporting	Interactive scientific & web-based charts
Data Handling	Excellent — huge datasets, live connections	Good — best with Google data sources	Excellent — in-memory associative engine	Good — strong with Zoho apps & common DBs	Good — strong with Python/R dataframes
Interactivity	High — drill-downs, filters, actions	Moderate — filters & controls	High — associative selections	Moderate to High — drill-downs & filters	High — zoom, hover, pan out of the box
Customization	High, but within Tableau's framework	Limited to templates & connectors	Moderate — extensions available	Moderate — templates plus custom widgets	High — extensive chart & layout options
Learning Curve	Medium	Low	Medium	Low	Medium
Integration	Wide (databases, cloud, APIs)	Deep with Google Workspace/ Analytics	Wide (databases, cloud, APIs)	Deep with Zoho Suite; also databases & APIs	Python, R, JS, Jupyter, Dash

Which One Should You Choose?

- Choose Tableau if you need enterprise-scale BI dashboards with rich interactivity and don't mind the licensing cost.
- Choose Looker Studio if you live in the Google ecosystem and want free, shareable marketing reports.
- Choose Qlik Sense if your team values free-form, associative data exploration over guided drill-downs.
- Choose Zoho Analytics if you're an SMB already using Zoho apps and want fast, affordable, guided reporting.
- Choose Plotly if you're working in Python/R and want interactive charts without leaving your data science workflow.